SEF® Lean Version

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AUDIENCE

This document is intended for enterprise architects, service architects, business architects, information architects, infrastructure architects, integration architects, security architects, programme managers, and business directors that are interested in managing the complexity of their organisations efficiently and effectively.

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AIM

The aim of this document is to describe the SEF® Lean Version, which is part of the Secure Enterprise Framework® (SEF®).

PURPOSE

The purpose of this document is to propose the SEF® Lean Version as the Enterprise Architecture Framework (EAF) for organisations or for programmes of large organisations.

INTRODUCTION

This document consists of three parts. The first part articulates the challenges of transformation programmes and the attempt to overcome them using existing Enterprise Architecture Frameworks (EAFs). The second part describes the components of the SEF® Lean Version. The third part presents the business benefits for utilising the proposed framework.

1 TRANSFORMATION CHALLENGES

Transforming organisations has always been challenging, and in many cases transformation programmes have not achieved the desired outcomes and benefits. Whether the aim of such programmes has been to replace existing systems with new ones, or to improve business outcomes, they all have one common factor that is often overlooked and underestimated: *complexity*.

Complexity is generally used to characterise something with many parts, where those parts interact with each other in multiple ways, culminating in a higher order of emergence[1], which is greater than the sum of its parts (Bertram and Van Dooble, 2019). This something is nothing other than a *complex system* and the study of these interactions at various levels is the main objective of *complex systems theory*.

A system is a group of interacting and interrelated parts that form a unified whole (Kim, 1999), whereas a *complex system* is a system with a behaviour intrinsically difficult to model due to the dependencies, competitions, relationships, or other types of interactions between its parts or between itself and its environment (Bertram and Van Dooble, 2019).

Although inherently an organisation exhibits the properties of a complex system, due to the embedded human factor, an organisation also shows the property of adaptation, which makes it a *complex adaptive system*.

A complex adaptive system is a system in which the unambiguous understanding of its individual parts does not automatically lead to the concise interpretation of its behaviour (Miller and Scott, 2007), and hence, the whole is more complex than its parts (Holland, 1998).

[1] In systems theory, emergence occurs when an entity is observed to have properties its parts do not have on their own

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Due to the inherent complexity in organisations, transformation programmes need an effortless way for managing this complexity and consequently freeing their resources to focus on how to achieve the outcomes and benefits.

Over the last thirty years, organisations have tried to effortlessly manage complexity through the utilisation of Enterprise Architecture Frameworks (EAFs). However, in many cases the use of EAFs has not assisted organisations to manage complexity. Instead, the introduction of complex concepts and practices has led to the addition of a complex layer to manage the complexity itself.

Ironically, from the business architecture perspective, managing complexity through a complex layer is an obvious sign of creating waste within the corresponding business processes. Such business processes are worthless because despite the herculean efforts delivered by resources, value cannot be added to the organisation. As a result, critics have constantly questioned the value of such frameworks and it is no coincidence that within the world of Enterprise Architecture, a character called *Zasyphus*[2] emerged to describe the corresponding distress of enterprise architects when dealing with this layer.

Unfortunately, it is human nature to overcomplicate the simple or to oversimplify the complex. By introducing relatively straightforward concepts but using excessively complex descriptions and methods has led to the addition of another layer of complexity. Some organisations have attempted to remove this layer by borrowing principles from the agile manifesto, which covers the delivery of software systems, not their development. Consequently, tailoring these principles to organisations has not been straightforward.

Regrettably, easy answers might have great appeal but they tend to cause further problems, because these answers are not close to the right ones. This oversimplification has resulted in the replacement of the complex layer by a vacuum that promotes anarchy and confusion. A typical example is the replacement of eternal designs by eternal nothingness. Perhaps, it is time to step back in a spirit of reflective and mature practice and rethink the way organisations and their parts should be modelled.

2 SEF® Lean Version

Although complexity in organisations should be managed using *simplicity*, this simplicity cannot be achieved by simplifying complex adaptive systems. Instead, the properties of these systems need to be recognised and the right level of transparency provided to observe them. This level of transparency can be obtained using the Secure Enterprise Framework® (SEF®).

SEF® considers enterprise architecture as a metaphor to describe *systems engineering* using business terms. Systems engineering is a transdisciplinary and integrative approach, based

[2] Zasyphus is a character that illustrates the suffering of Sisyphus within the context of Enterprise Architecture. Before Enterprise Architects finish modelling a system, the state of this system changes (i.e., needing remodelling before the system is modelled).

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on systems thinking[3], that concentrates on designing, integrating, and managing complex systems over their lifecycles, leading to the successful realisation, use, and retirement of *engineered systems*[4] (INCOSE).

In line with the above consideration, SEF® has simplified the enterprise architecture framework. It does this by narrowing the distance between enterprise architecture and systems engineering, by keeping a reasonable number of practical concepts and components, by discarding those that cannot add value, and by ensuring that the scope of the framework covers the organisation as a whole rather than selected functions (e.g., Information Technology).

In order to cover organisations and programmes of large organisations, SEF® has been simplified to shorten the time needed for its introduction and utilisation. The result of this simplification is the SEF® Lean Version, which is a subset of SEF®.

1.1 Overview

The SEF® Lean Version, illustrated in fig. 1, uses the ***model*** to provide an approximate representation of the ***system***, which in turn, is an organised entity composed of interrelated and interdependent parts.

Models are used to represent existing systems, through their current state, and to develop new ones through their target state. An existing system is denoted by the ***current state***, whereas the new system is denoted by the ***target state***. Any state between current and target is called ***interim state***. The purpose of interim states is to facilitate the transition from the current state towards the target state.

The model, by which any state is represented, is composed of a set of ***domains*** that are used to describe the aspects of the system. This set comprises the **offering domain**, ***business domain, knowledge domain, application domain, infrastructure domain,*** and the ***facility domain***. The model also consists of ***profiles***, which cover the domains of the model. These profiles are the ***security profile*** and the ***interface profile***. Domains and profiles are described separately.

The model is described by three ***architectures***: ***conceptual, behavioural,*** and ***implementation***. These architectures are composed of ***building blocks***. Each of these blocks represents a concept with a defined ***boundary*** that delivers a finite ***function***. This function can be delivered individually or in collaboration with adjacent building blocks through a set of defined ***interfaces***. These interfaces can be ***internal*** or ***external***.

The conceptual architecture defines ***WHAT*** building blocks form the model. The behavioural architecture describes ***HOW*** these building blocks interoperate to deliver the expected behaviour of the model. The implementation architecture specifies ***WITH WHAT*** solutions

[3] Systems thinking is a holistic analytic approach that focuses on how the constituent parts of a system interrelate, and how it works over time alone and within the context of larger systems.

[4] An engineered system is a system designed or adapted to interact with an anticipated operational environment to achieve one or more intended purposes while complying with applicable constraints.

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the defined building blocks produce the behaviour of the model. Only these identified solutions become part of the system through the corresponding states (i.e., current, interim, target).

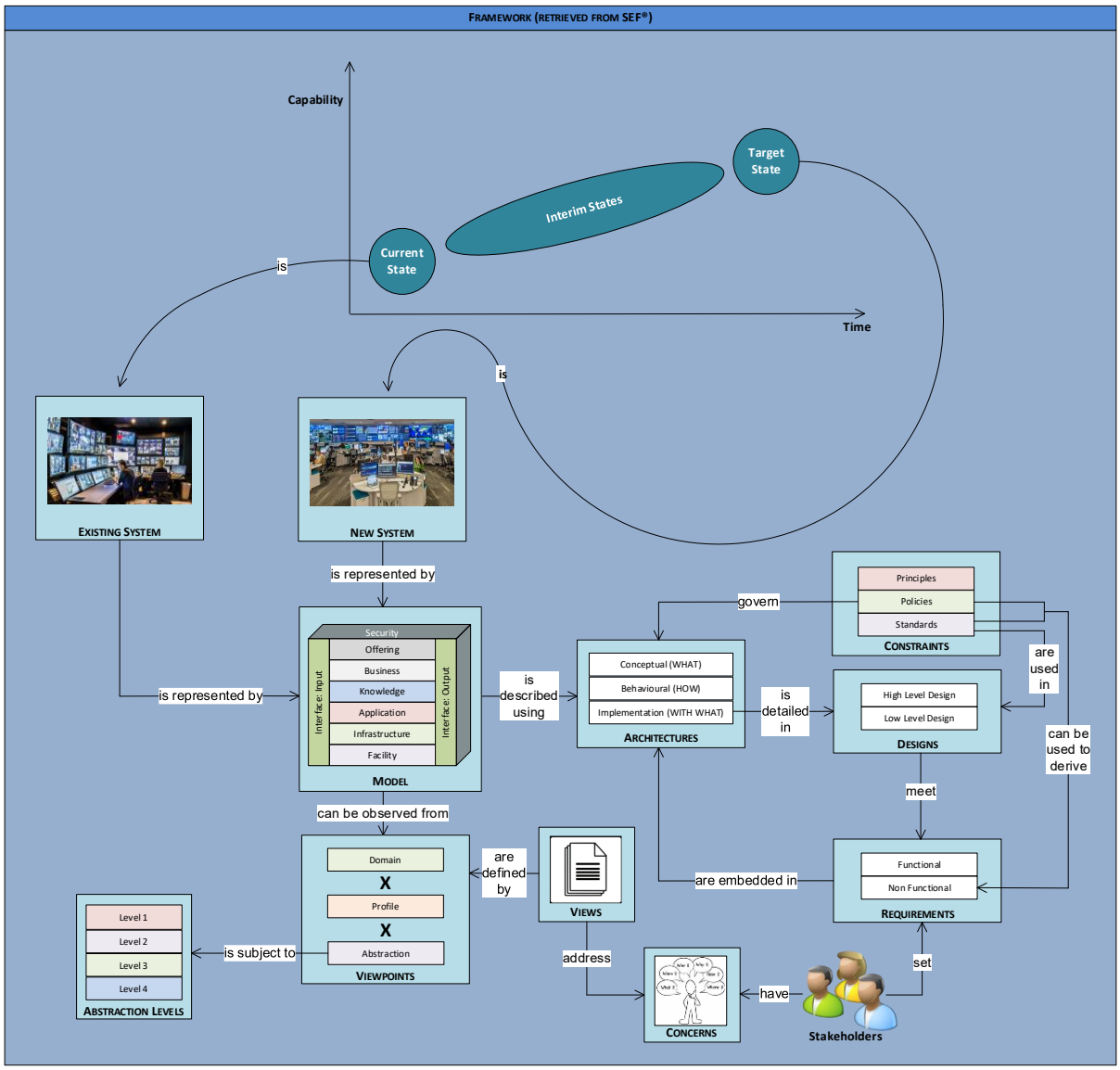

Figure 1. SEF ® Lean Version: Overview

Solutions identified in the implementation architecture may be represented by ***products***. Products are applicable to the application and infrastructure domains (e.g., a web application server running on an operating system). However, some of these products may be addressed through a profile (e.g., a privileged session manager addressed through the security profile). Any selected solution, including products, meets the ***requirements*** set by ***stakeholders***. This solution is presented in a ***high-level design*** and, at least, in a ***low-level design***. This implies that the relationship between a high-level design and a low-level design is ***one to many***. The former describes the components of the solution and how these interoperate with adjacent solutions through defined interfaces. The latter describes the details of how these components are deployed and utilised.

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Architectures are governed by a set of ***constraints***. These constraints are ***principles***, ***policies***, and ***standards***. Principles are **norms**, they are used to achieve **consensus**, and are enforced by policies. These policies are **rules**, they are used to achieve **order**, and are supported by standards. In turn, standards are **specifications** and are used to achieved **uniformity**.

Architectures have requirements embedded in them. These requirements are split between ***functional*** and ***non-functional requirements***. Although both categories are produced by stakeholders, non-functional requirements acting as constraints are derived from corresponding policies and standards. Within the model, all requirements flow down from the offering domain to the facility domain.

Likewise, within the description of the model, all requirements flow down from the conceptual architecture to the corresponding low-level design. Consequently, all the solutions, together with their components, interoperations, interfaces, and configurations, comply with policies and standards, which ensure uniformity and standardisation within the ***enterprise landscape***.

Stakeholders who have an ***interest*** in the system also have ***concerns***. These concerns are addressed through selected ***views***, which are defined by ***viewpoints***. Each viewpoint defines the way the model can be seen through a defined level of ***abstraction*** in a given intersection between domain and profile. Abstractions are defined by levels, which are standardised in the whole model (e.g., Level 1 in the offering domain corresponds to Level 1 in the business domain). Summarising the view and viewpoint relationship, every view (e.g., document) has an associated viewpoint (e.g., document template) that describes it.

1.2 Reference Model

In order to assure that models under development follow industry ***good practices***, the SEF® Lean Version utilises ***design principles*** and ***design patterns*** to drive the development of ***reference architectures***. These architectures comprise conceptual and behavioural architectures.

The reference architecture is utilised to describe a ***reference model***, which can then be tailored to produce the model for a specific system. In this way, a reference architecture can be used to develop the corresponding architectures of the model, while good practices are inherited to produce the necessary constraints needed to govern these architectures. This relationship is illustrated in fig. 2.

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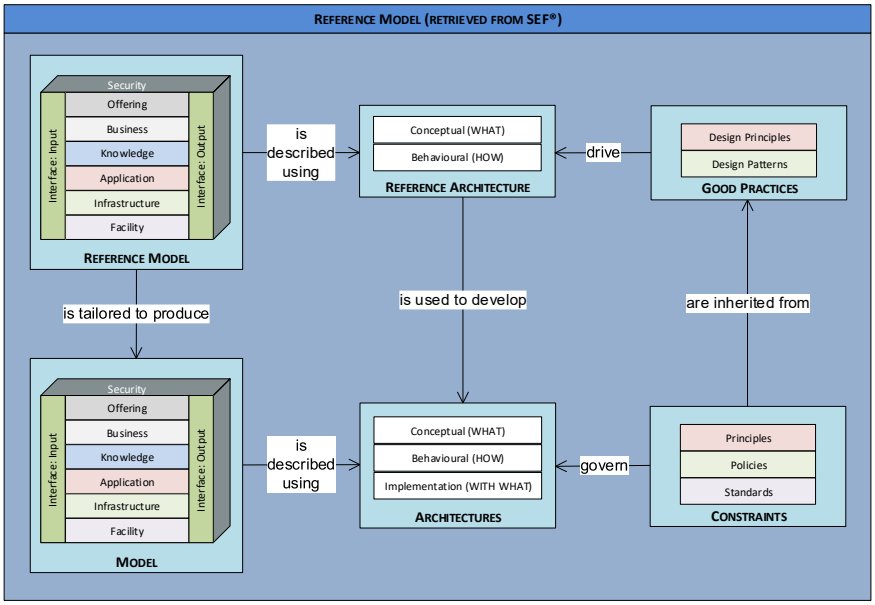

Figure 2. SEF® Lean Version: Reference Model

1.3 DOMAINS

The SEF® Lean Version provides a set of domains to describe the aspects of the system that needs modelling. Within the model, all requirements follow a top-down approach to ensure they always deliver business benefits. These domains, depicted in fig. 3, provide a layered stack of interconnected and interoperating building blocks that are needed to deliver the expected outputs and outcomes. There are six domains in a model: ***offering***, ***business***, ***knowledge***, ***application***, ***infrastructure***, and ***facility***.

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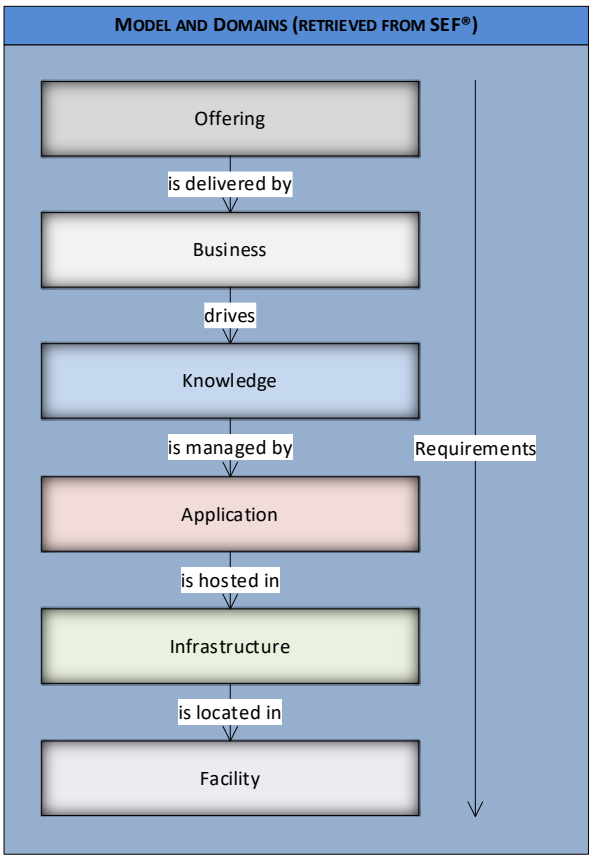

Figure 3. SEF® Lean Version: Domains

1.3.1 OFFERING DOMAIN

The offering domain contains the building blocks that describe the products of the model. These products can be goods, services, or a combination of both. Manufacturing is an example of a vertical market that offers goods, whereas finance is an example of a vertical market that offers services.

Services are further divided into ***business services*** and ***supporting services***, where the former are the services delivered by the model, whereas the latter are the services that support the former. It is important to distinguish these types of services because the business services are consumed externally by another model, while the supporting services are consumed internally. Some of these services might be offered by another model, which makes them subject to outsourcing or offshoring.

1.3.2 BUSINESS DOMAIN

The business domain contains the building blocks that deliver the products of the offering domain. These building blocks represent the business processes and include process maps, roles, rules, and procedures.

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1.3.3 Knowledge Domain

The knowledge domain is driven by the business domain and its building blocks support the execution of the business processes of the business domain. These building blocks represent knowledge[5], information, and data and include lexicons, taxonomies, metamodels, lifecycles, classifications, and presentations.

1.3.4 Application Domain

The application domain contains the building blocks to manage the explicit knowledge of the knowledge domain and to support the business processes of the business domain. These building blocks represent applications and include software solutions, tools, SaaS offerings, middleware, and data management (e.g., Relational Database Management System). In this domain, when business processes are automated, they are executed in the form of software processes or threads.

1.3.5 Infrastructure Domain

The infrastructure domain contains the building blocks that represent the physical and virtual infrastructure where the application building blocks are hosted. Hypervisors, operating systems, devices, and networks are described in this domain. In some cases, this infrastructure is abstracted under the public cloud services, including IaaS and PaaS offerings.

1.3.6 Facility Domain

The facility domain contains the building blocks that represent the physical buildings that host the infrastructure and people. These physical buildings are split into two categories, the frontend that covers offices, operation centres, and homes, and the backend that covers data centres.

1.4 Profiles

The SEF® Lean Version provides a set of profiles, which cover all the domains of the model, but they describe these domains differently. When these profiles are combined, domains are described in full. Although SEF® utilises a wide range of profiles, this lean version covers only two: ***interface*** and ***security***. These profiles are depicted in fig. 4.

[5] Knowledge is categorised as tacit and explicit. The former represents the know-how and learning, which is embedded within the minds of people. The latter represents distilled and codified information. This information is processed, organised, and structured data, which is presented in a given context. Data comprise raw and unorganised facts.

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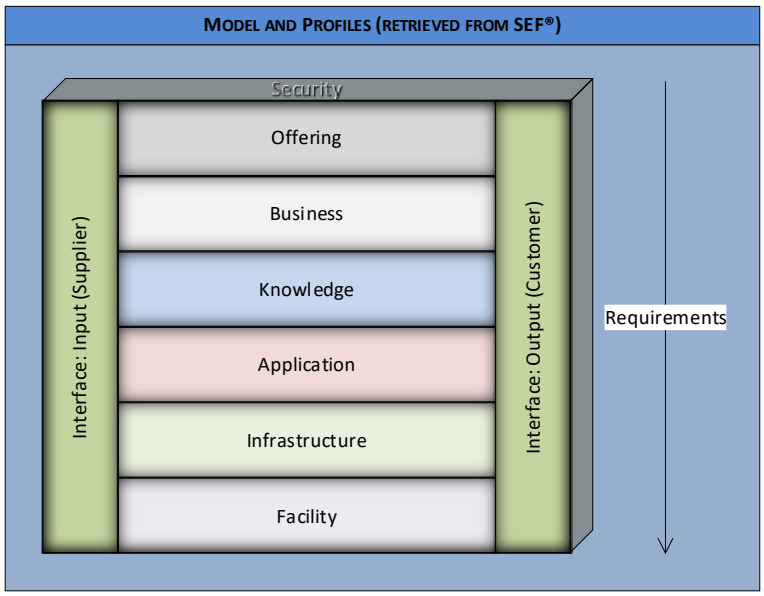

Figure 4. SEF® Lean Version: Profiles

1.4.1 INTERFACE PROFILE

The interface profile describes the building blocks that represent the interfaces and data flows between the domains. These building blocks cover the interoperation between the domains of a model, and is defined as the ***inter-domain collaboration***. The interoperation between the domains of different models is defined as the ***inter-model collaboration***. The former collaboration applies to a system, whereas the latter applies to the supply chain of this system. When the interface profile applies to the supply chain, this profile is divided into ***input*** and ***output*** to support ***suppliers*** and ***customers***. This support is illustrated in fig. 5.

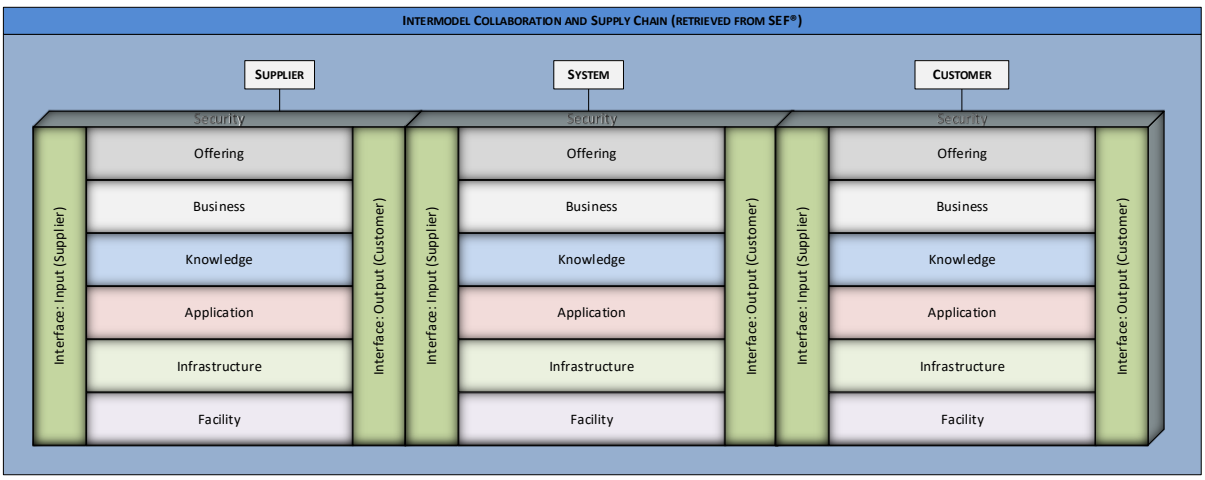

Figure 5. SEF® Lean Version: Supply Chain

1.4.2 SECURITY PROFILE

The security profile, depicted in fig. 6, describes the building blocks that cover the ***threat profile***, the ***security risk profile***, and the ***security control profile*** of the model. These three profiles are closely linked. The threat profile identifies and assesses the threats to the

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assets. The security risk profile calculates the probable manifestation of the threats and the corresponding impact they might cause to assets. The security control profile devises the corresponding security controls that are needed to reduce the likelihood of the manifestation and to contain the potential impact.

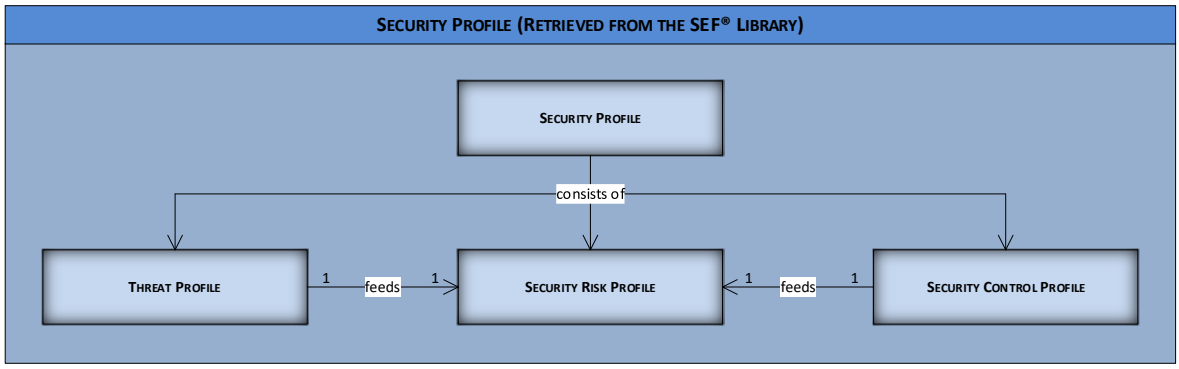

Figure 6. SEF® Lean Version: Security Profile

More specifically, the threat profile supplies the security risk profile with threats and weaknesses, which in turn, become the inherent security risks and issues of the security risk profile. Meanwhile, the security control profile feeds the security risk profile with deterrent, preventive, detective, and corrective controls, which assist the security risk profile to deduct the residual security risks from the already identified inherent security risks.

1.5 MODELLING: MODELS, DOMAINS, PROFILES

The SEF® Lean Version acknowledges that organisations operate in the market as *complex adaptive systems*, which learn and adapt to the changes in the environment. Those organisations that fail to learn and adapt to changes in the market cease to exist. As stated by Obolensky (2014), in order for organisations to adapt to external changes, they need to have a high degree of awareness of their surroundings, as well as a high capability to change internally. This principle flows down to the smallest part of the organisation. The SEF® Lean Version therefore strongly recommends, before concentrating on the system itself, to focus on its supply chain, upstream and downstream.

Modelling the supply chain of a system requires modelling the corresponding models, domains, and profiles. This practice requires the use of models for domains and profiles that are managed by business functions. For example, consider a simple scenario, illustrated in fig. 7, that consists of an organisation with a security business function and an IT business function. The former function operates within the organisation, whereas the latter function is split between in-house and outsourced to a service provider.

In this scenario, the in-house IT function provides IT services, under the offering domain, to manage the application and infrastructure domains of the organisation. These services are delivered by corresponding business processes covered by the business domain, which in turn uses explicit knowledge, information, and data that are covered by the knowledge domain. Paradoxically, the IT function needs applications and infrastructure to support the management of data and to support its processes. This infrastructure is located in a specific location under the facility domain. For some reason, the organisation decides to outsource

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some of the IT services to a service provider. This service provider becomes the supplier of IT services to the in-house IT function, which in turn, repackages these services to the organisation. In turn, this supplier needs the corresponding business processes to deliver these services and the remaining domains and profiles to support them.

Based on the same scenario, since the security function is provided by the organisation, all the security services are covered by the offering domain of this model. This function has its own business processes to deliver its services and needs specific explicit knowledge, information, and data to support them. However, the application and infrastructure domains are managed by the IT function.

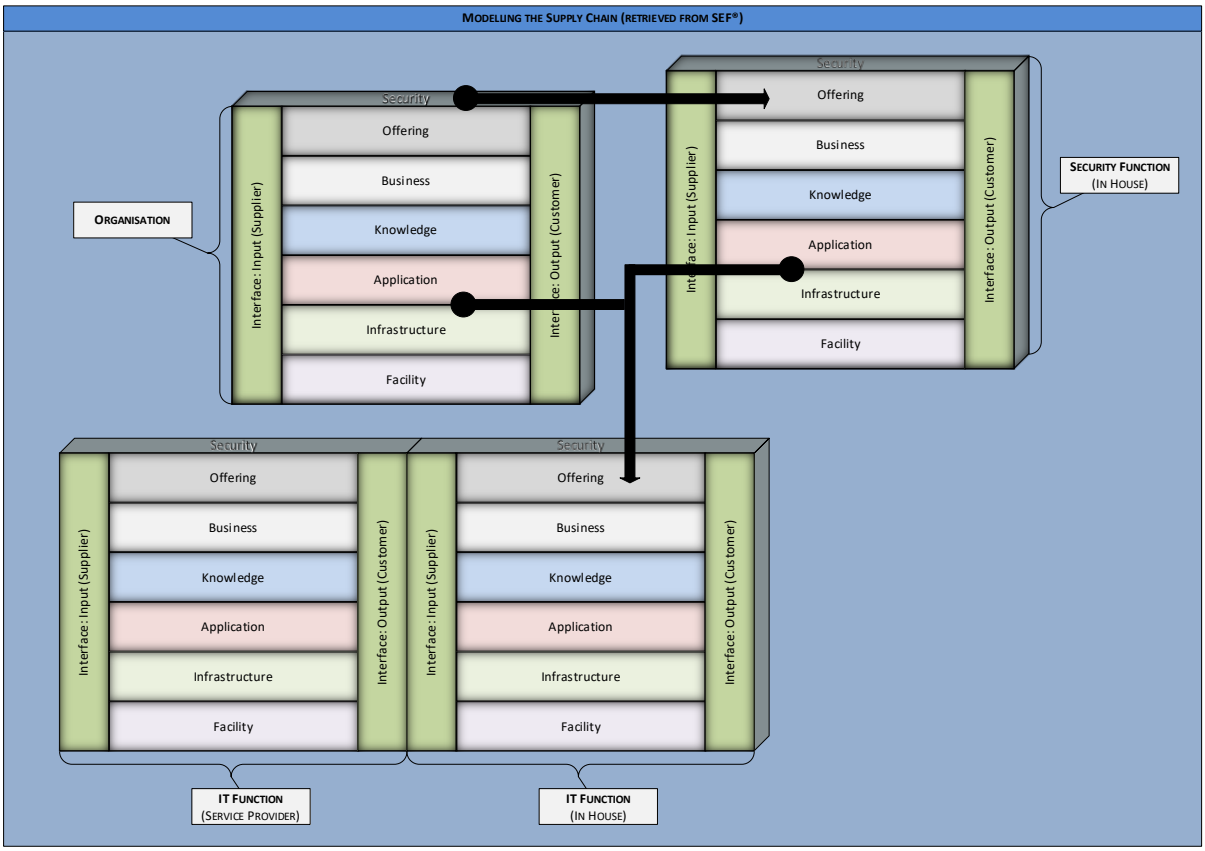

Figure 7. SEF® Lean Version: Modelling

1.6 VIEWS AND VIEWPOINTS

In order to address the concerns of stakeholders, the SEF® Lean Version (like many EAFs) recommends the utilisation of views to describe the model from a viewpoint that is common to a group of stakeholders. Unlike common perceptions, views are decoupled from architectures and designs for the simple reason that most stakeholders are not interested in them.

For example, a Payment Card Industry Data Security Standard (PCI DSS) Compliance Manager is not interested in any of the architectures (i.e., conceptual, behavioural, and implementation) or designs (i.e., high level and low level). This stakeholder is interested to know what security controls are in place to address the PCI DSS requirements and where these controls are described. Therefore, the concern of this stakeholder is limited to

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compliance with PCI DSS and the view of the model should address it. This view can be shared with internal and external auditors, who perform PCI DSS audits. The corresponding viewpoint should drive the production of the view using a specific document template, report template, or script, or a combination of these three.

Before producing views, stakeholders need to agree on the specification of their corresponding viewpoints. That is, they need to agree on the scope of the view by deciding on the domains that need to be included, the profiles that need to be considered, and the level of abstraction the view needs to focus on. These three filters, namely domain, profile, and abstraction, which are depicted in fig. 8, should be unambiguously recorded in the viewpoint (e.g., template).

Apart from the scope, it is important to define the format in which to present the information. This format should also include the type of language used. For example, a view for a business stakeholder needs to be free of technical terms. If some of these terms are unavoidable, a link to the lexicon (i.e., dictionary) should be included. In regard to the contents, the information in the view should be enough to address the concerns of the stakeholders. Providing more or less information than required could lead to extra time being spent on addressing misunderstanding and confusion, which could have otherwise been avoided.

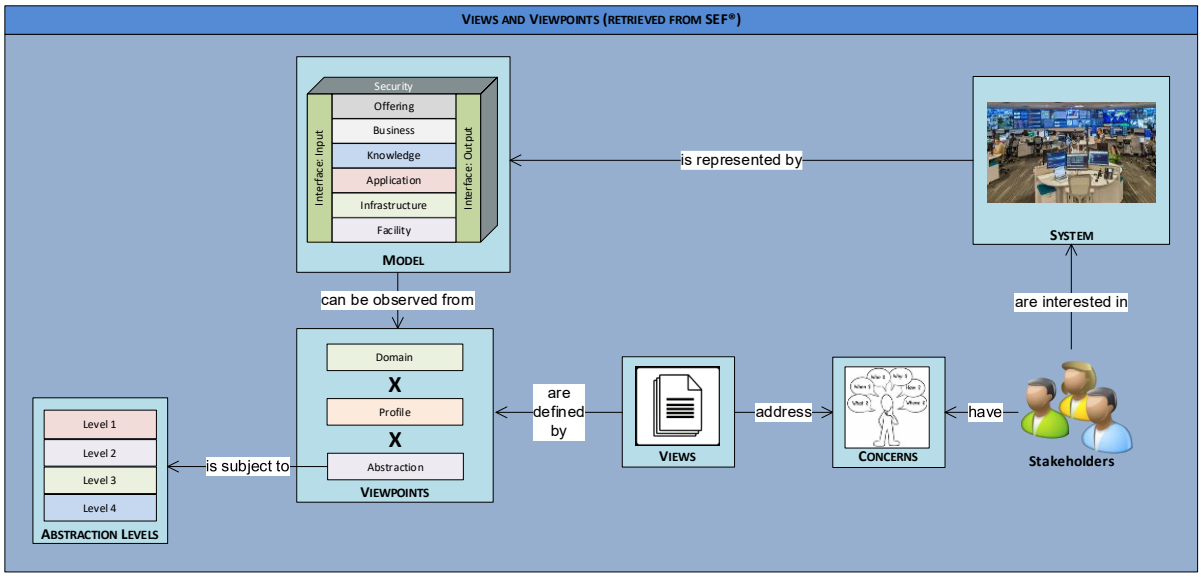

Figure 8. SEF® Lean Version: Views and Viewpoints

1.7 GOVERNANCE

The governance of the SEF® Lean Version, described by its metamodel and depicted in fig. 9, centres around the concept of the ***Design Authority***. The aim of this authority is to protect the enterprise landscape against ***risks*** introduced by models. The introduction of such risks could undermine the function, mission, compliance, trust, or liquidity of the organisation. For example, a model may solve a problem but also create a few more, or it may not solve a problem, but shift it somewhere else. These problems could destabilise a specific function of the organisation, or its entire mission. In addition, they could cause or foster compliance or security issues, which in turn, could impact the trust of suppliers and customers. These

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problems could also cause a financial impact, which may affect the immediate liquidity of the organisation.

The Design Authority consists of ten ***tiers***, where the first six are internal and the remaining four are external. These tiers are described separately.

This Design Authority is governed by its ***Terms of Reference (TOR)*** and is managed by the ***Board***, which consists of eight ***members***. One of these members is the ***Chair***, and is represented by the ***Chief Enterprise Architect***, who covers the model. The remaining members cover their corresponding domains and profiles of the model.

More specifically, the ***Enterprise Service Architect*** covers the offering domain, the ***Enterprise Business Architect*** covers the business domain, the ***Enterprise Information Architect*** covers the knowledge domain, the ***Enterprise Application Architect*** covers the application domain, the ***Enterprise Infrastructure Architect*** covers the infrastructure and facility domains, the ***Enterprise Integration Architect*** covers the integration profile, and the ***Enterprise Security Architect*** covers the security profile. It is worth noting that these roles should be undertaken by people who have the corresponding knowledge and skills.

The Board of the Design Authority schedules ***Design Reviews***, which are ***Compliance Reviews***. During these sessions, the Board reviews ***deliverables*** submitted by ***architects***, ***engineers***, and ***modellers***, to ensure they comply with the corresponding ***constraints*** maintained by the Board. Constraints and deliverables are published and stored in the ***enterprise repository*** respectively.

The Design Authority always operates at ***level 1***, which covers the organisation. Optionally, this authority may operate at ***level 2***, covering a programme of the organisation. At level 1, the authority is permanent, whereas at level 2 the authority is temporary and its operation ends when the programme finishes.

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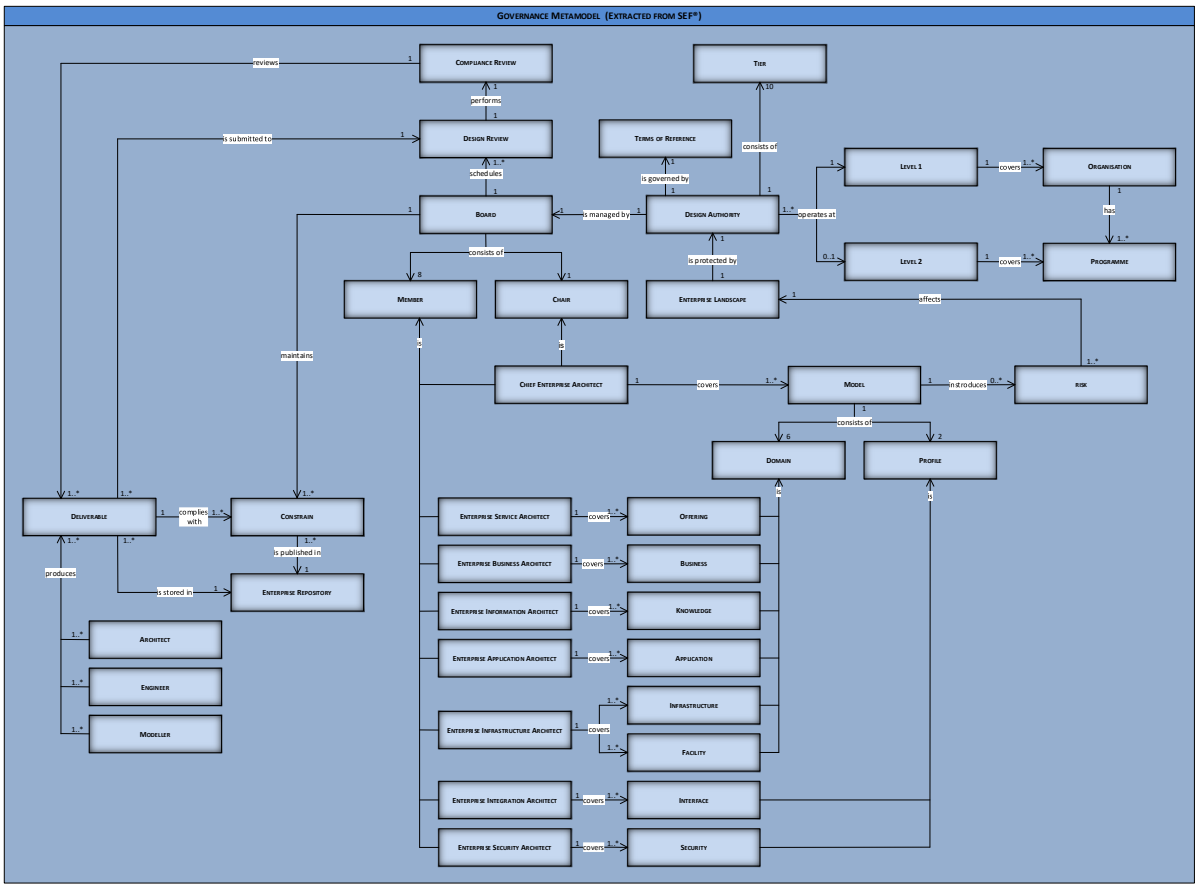

Figure 9. SEF® Lean Version: Governance Metamodel

1.7.1 Governance Levels

The SEF® Lean Version recommends organisations decide on the operational model of their design authorities before implementing them. This operational model is defined using two levels: ***level 1*** and ***level 2***.

Regardless of the size of these organisations, all design authorities should operate at Level 1. This level is depicted in fig. 10. Internally, at this level all programmes of the organisation are covered by the same Organisation Design Authority. Externally, an Organisation Design Authority aligns with the corresponding design authorities of supplier and customer organisations. This alignment should be limited to the relevant constraints that fall within the scope of the supply chain only. That is, the alignment should be limited only to policies and standards that support the collaboration between organisations. Unlike common perceptions, this alignment should not cover the whole range of policies and standards utilised by organisations. Likewise, principles and Enterprise Architecture Frameworks (EAFs) should remain out of scope.

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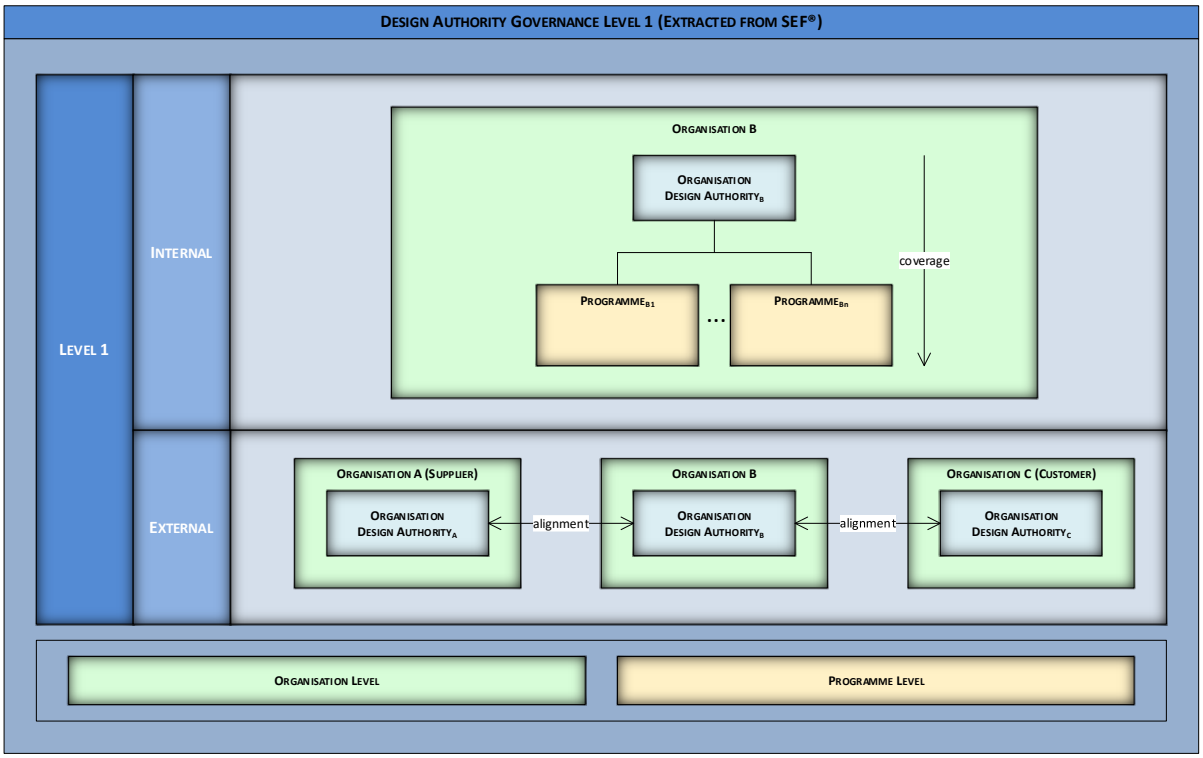

Figure 10. SEF® Lean Version: Design Authority – Governance Level 1

When large organisations run programmes of considerable size, the SEF® Lean Version recommends the use of design authorities operating at level 2. This level is depicted in fig. 11. At this level, a temporary Programme Design Authority is embedded in the programme throughout its duration. The Programme Design Authority inherits all the constraints from the Organisation Design Authority and represents the latter at the programme level.

Within the programme, a Programme Design Authority collaborates with the Programme Delivery Authority to ensure the deliverables solve the problems without shifting them somewhere else or creating new ones, they deliver the expected value to the organisation without creating debt, and are delivered within their corresponding timeframe and budget. That is, while the role of the Programme Delivery Authority is to deliver the programme to the agreed scope, budget and timescale, the role of the Programme Design Authority is to select a set of integrated solutions that deliver the expected value without introducing risks to the enterprise landscape.

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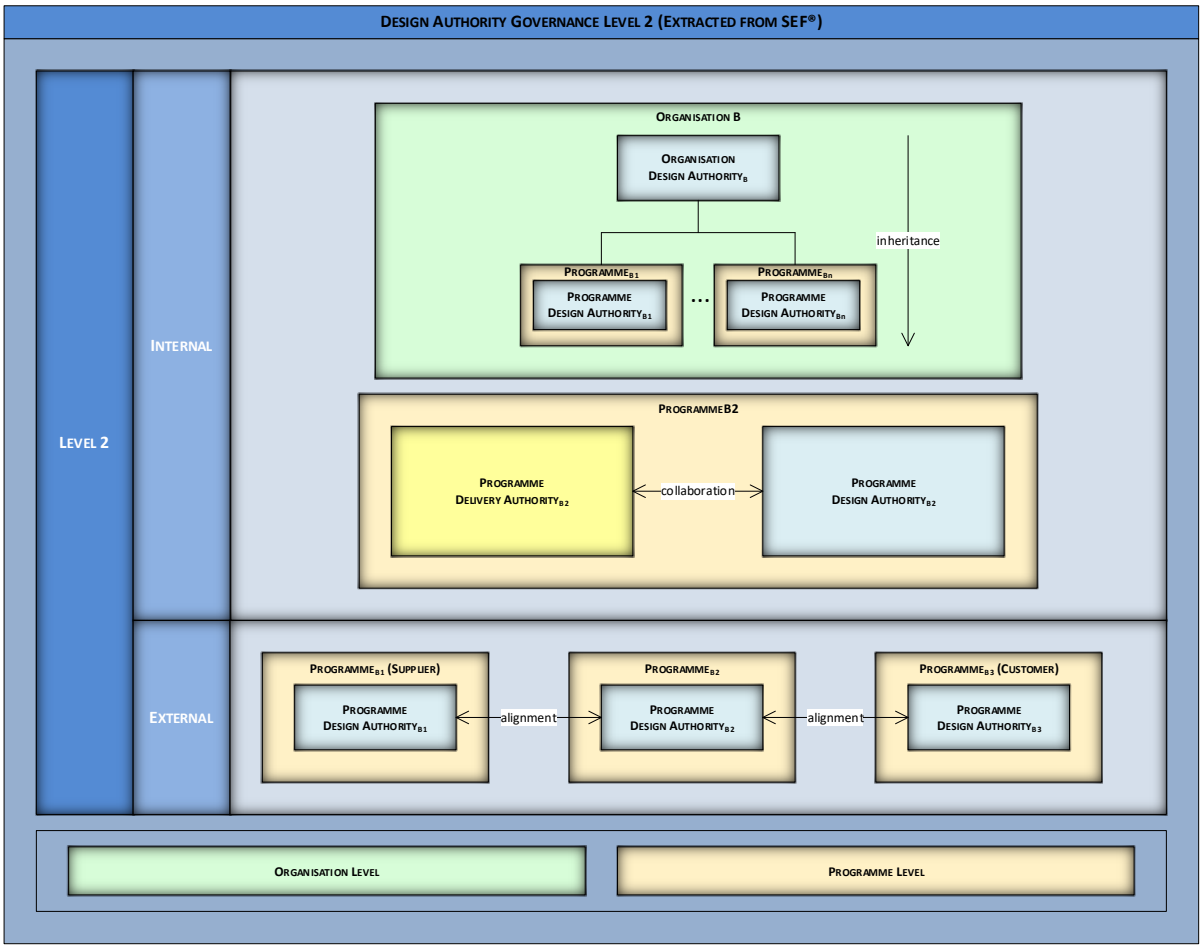

Figure 11. SEF® Lean Version: Design Authority – Governance Level 2

When programmes run in parallel, these temporary design authorities align themselves to ensure that all programmes deliver the intended values through the selected solutions and do not introduce unacceptable and unmanageable risks in the enterprise landscape of the organisation.

Since all these authorities utilise the same constraints, this alignment is focused on integration and risk mitigation. Therefore, ensuring that all solutions add the required values to the organisation without leaving gaps behind that could impact the corresponding business processes, and without creating problems in other areas that are not covered by the same programme.

1.7.2 GOVERNANCE TIERS

The structure of the Design Authority is based on ten governance tiers. The first six tiers are internal and the remaining four are external. These tiers are depicted in fig. 12 and described in table 1.

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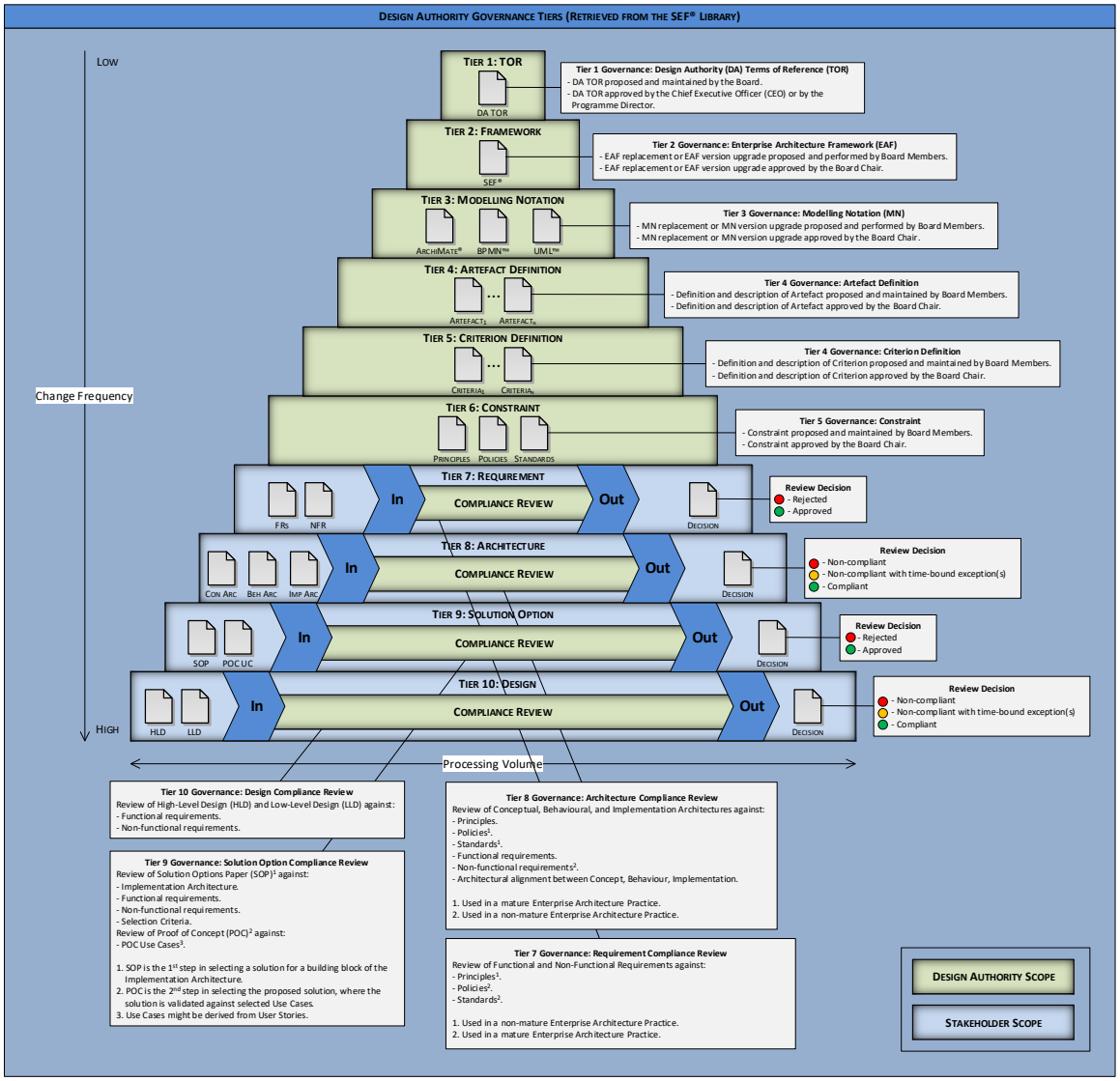

Figure 12. SEF® Lean Version: Design Authority – Governance Tiers

Tier			
Scope	**Level**	**Name**	**Description**
Internal	1	Terms of Reference	This governance tier focuses on the Terms of Reference (TOR) of the Design Authority (DA). These terms are proposed and maintained by the Board. Depending on the level where the DA operates, these terms are approved by the Chief Executive Officer (CEO) or by the Programme Director.
	2	Enterprise Architecture Framework	This governance tier focuses on the Enterprise Architecture Framework (EAF). The replacement of the existing EAF by another different one, or the replacement of the current EAF version by a newer version of the same EAF is proposed and performed by Board Members and approved by the Board Chair.

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Tier			
Scope	**Level**	**Name**	**Description**
	3	Modelling Notation	This governance tier focuses on the Modelling Notations (MNs) that are utilised to describe the model. The SEF® Lean Version strongly recommends the use of the latest standard notations utilised by industry. These modelling notations may include ArchiMate®, BPMN™, and UML™. Legacy modelling languages or legacy notations are not recommended. The replacement of an existing MN by another different one, or the replacement of a current MN version by a newer version of the same MN is proposed and performed by Board Members and approved by the Board Chair.
	4	Artefact Definition	This governance tier focuses on defining the types of artefacts needed to view the model, and on describing their templates. These templates are known as viewpoints. In addition, this tier focuses on describing the architecture and design templates. The definition and description of artefacts are proposed and maintained by Board Members, and are approved by the Board Chair.
	5	Criterion Definition	This governance tier focuses on defining and describing the criteria needed to select the solutions of the implementation architecture. These criteria may cover the areas of functionality, integration, support, performance, and Total Cost of Ownership (TCO). The definition and description of criteria are proposed and maintained by Board Members, and are approved by the Board Chair.
	6	Constraint	This governance tier focuses on the constraints of the model. These are principles, policies, and standards. These constraints are proposed and maintained by Board Members, and are approved by the Board Chair.
External	7	Requirement	This governance tier focuses on performing compliance reviews of Functional and Non-Functional Requirements against: - Principles. - Policies. - Standards. These reviews are delivered through Design Review sessions, by the end of which, the proposed requirements are either Approved or Rejected by the Board. Rejected requirements could be reviewed again if their non-compliant parts are addressed. Please note that Policies and Standards are used in mature Enterprise Architecture Practices.

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Tier			
Scope	Level	Name	Description
	8	Architecture	This governance tier focuses on performing compliance reviews of Conceptual, Behavioural, and Implementation Architectures against: - Principles. - Policies. - Standards. - Functional requirements. - Non-functional requirements. - Architectural alignment between Concept, Behaviour, Implementation. These reviews are delivered through Design Review sessions, by the end of which, the Board decides if the proposed architectures are: - Non-compliant - Non-compliant with time-bound exception(s) - Compliant Please note that Policies and Standards are used in mature Enterprise Architecture Practices.
	9	Solution Option	This governance tier focuses on performing compliance reviews of Solution Options Papers (SOPs) against: - Implementation Architecture. - Functional requirements. - Non-functional requirements. - Selection Criteria. These reviews are delivered through Design Review sessions, by the end of which, the proposed solution options are either Approved or Rejected by the Board. Approved solutions might be assessed further using the Proof of Concept (POC) evaluation method. When evaluating solutions using POCs, their POC is subject to compliance reviews against POC Use Cases. These Use Cases might be derived from User Stories. By the end of the corresponding Design Review session, the proposed solution is either Approved or Rejected by the Board.
	10	Design Compliance Review	This governance tier focuses on performing compliance reviews of High-Level Design (HLD) and Low-Level Design (LLD) against: - Functional requirements. - Non-functional requirements. These reviews are delivered through Design Review sessions, by the end of which, the Board decides if the proposed designs are: - Non-compliant - Non-compliant with time-bound exception(s) - Compliant

Table 1. SEF® Lean Version: Design Authority – Description of Governance Tiers

3 Benefits

The SEF® Lean Version provides the following business benefits:

A. **Fast Deployment**. Organisations and programmes of large organisations can setup the SEF® Lean Version in a matter of days. Reasonable time may be needed to develop the constraints and the corresponding artefacts. It is strongly recommended to start with the Terms of Reference (TOR) of the Design Authority, a small set of principles, and a few templates needed for the architectures, designs, and views.

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B. **Effortless Learning**. The SEF® Lean Version is described in twenty-three pages, uses common engineering and enterprise architecture terms, and recommends the utilisation of industry standards for modelling.

C. **Improved Communication**. The SEF® Lean Version uses concise terms and an unambiguous structured method for modelling systems and addressing the concerns of stakeholders. In addition, it recommends the utilisation of common standard notations as a way to develop a common language with all stakeholders to avoid misunderstandings and to improve their communication. Furthermore, by providing all the relevant information of the enterprise landscape in an accessible format, through the enterprise repository, stakeholders can access the information they require whenever the need arises.

D. **Enhanced Collaboration**. The SEF® Lean Version assists in breaking down monolithic and hierarchical structures that obstruct and complicate the collaboration between stakeholders. This collaboration is supported by a robust governance that starts at the programme level, continues at the organisation level, and extends to the supply chain.

E. **Better Standardisation**. The SEF® Lean Version fosters a high level of standardisation within the enterprise landscape. This standardisation supports higher integration with less efforts and costs, reduced duplication and redundancies, optimal flexibility, and improved business productivity.

F. **Complexity Management**. It is impossible to manage something that cannot be observed or even seen. The SEF® Lean Version sets the foundations for obtaining the transparency needed to manage complexity. Managing complexity effortlessly leads to more efficient business operations with lower costs, more shared capabilities, and lower management costs.

G. **Increased Agility**. In the era of rapidly evolving technology and often disruptive transformation, the need for agility is imperative. The use of the SEF® Lean Version provides an organisation with increased agility. Organisations with a holistic view and understanding of their enterprise landscape are better equipped to react to external events quickly and uniformly. This increased agility provides organisations with the competitive advantage to exploit opportunities at the right time and address threats before they are manifested.

H. **Reduced Total Cost of Ownership (TCO).** The SEF® Lean Version enables better and faster decision making, more efficient resource allocation, leveraging shared capabilities, and the reduction of operational costs. As an open source EAF, combined with fast deployment and effortless learning, this lean version of SEF® amounts to elevated Return on Investment (ROI) and promises an admirable TCO.

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GLOSSARY

Glossary	
Term	**Description**
BPMN	Business Process Modelling and Notation
CC	Creative Commons
EAF	Enterprise Architecture Framework
IaaS	Infrastructure as a Service
PaaS	Platform as a Service
PCI DSS	Payment Card Industry Data Security Standard
ROI	Return on Investment
RDBMS	Relational Database Management System
SaaS	Software as a Service
SEF	Secure Enterprise Framework
TCO	Total Cost of Ownership
UML	Unified Modelling Language

REFERENCES

BERTRAM, L., VAN DOOBLE, G. (2019). *Nomenclatura - Encyclopedia of Modern Cryptography and Internet Security*. Books on Demand.

HOLLAND, J. (1998). *Emergence: From Chaos to Order*. Perseus Books Group.

INCOSE. *Systems and SE Definitions* [ONLINE]. INCOSE. Available from https://www.incose.org/about-systems-engineering/system-and-se-definition/system-and-se-definitions. [Accessed: 4 December 2019].

KIM, D. (1999). *Introduction to Systems Thinking*. Pegasus Communications, Inc.

MILLER, J., SCOTT, P. (2007). *Complex Adaptive Systems: An Introduction to Computational Models of Social Life*. Princeton University Press.

OBOLENSKY, N. (2014). *Complex Adaptive Leadership: Embracing Paradox and Uncertainty*. 2nd Edition. Farnham, Gower Publishing.

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